

Frame Field Generation

Problem Setup and Minimization on a Simple Domain

Structured Grid Generation via Frame Fields

December 11, 2025

Outline

1 Background

2 Optimization Problem

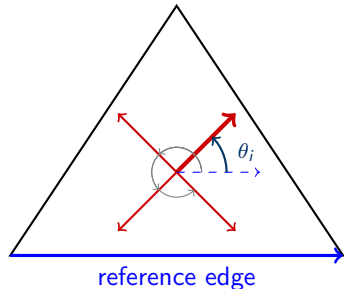
What is a Frame Field?

Definition:

- A frame field assigns a cross (4 orthogonal directions) to each vertex (or face) of a mesh
- Rotational symmetry: directions are equivalent up to $\pm\pi/2$ rotations
- Used to guide quadrilateral mesh generation

Angle Representation θ_i :

- Reference edge: local coordinate system
- Angle θ_i : rotation of frame
- 4 directions by rotating θ_i by $r \cdot \frac{\pi}{2}$, $r \in \{0, 1, 2, 3\}$.



Frame rotated by angle $\theta_i = 45^\circ$.

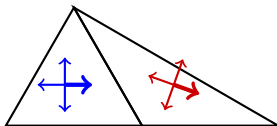
Goal: Smooth Frame Field

Objective

Find a frame field that is as *smooth* as possible across the mesh, while respecting a set of user constraints.

Smoothness means:

- Neighboring triangle faces f_i and f_j should have aligned frames
- Minimize rotation difference between adjacent faces
- Allow for integer period jumps ($\pi/2$ rotations)



Smooth transition between frames

Local $\pi/2$ Rotational Symmetry

Key observation:

- Cross frames have 4-fold rotational symmetry
- Locally rotating by $\pi/2$ gives an equivalent frame
- Adjacent frames can differ by integer multiples of $\pi/2$

Period jump p_{ij} :

- Integer value: $p_{ij} \in \mathbb{Z}$
- Represents number of $\pi/2$ rotations
- Allows frames to "jump"

Examples of period jumps:

$$p_{ij} = 0 \text{ (no jump)}$$



$$p_{ij} = 1 \text{ } (\pi/2 \text{ jump})$$



$$p_{ij} = 3 \text{ } (3\pi/2 \text{ jump})$$



Period jumps allow frames to align while respecting $\pi/2$ symmetry

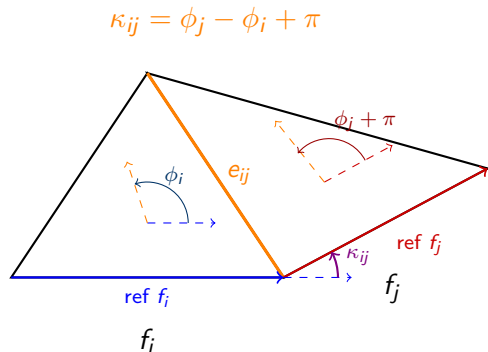
Transport Angle: κ_{ij}

Definition:

- $\kappa_{ij} \in (-\pi, \pi]$ is angle between local frame of f_i and f_j
- Depends only on triangle geometry (fixed)

Computation:

- 1 Identify shared edge e_{ij}
- 2 In f_i : compute angle ϕ_i between its reference edge and e_{ij}
- 3 In f_j : compute angle ϕ_j between its reference edge and e_{ij} (note orientation requires π addition)
- 4 Set $\kappa_{ij} = \phi_j - \phi_i + \pi$ (adjusted to $(-\pi, \pi]$)



Transport angle κ_{ij} is **fixed** (geometric property)

Coordinate Transformation Between Faces

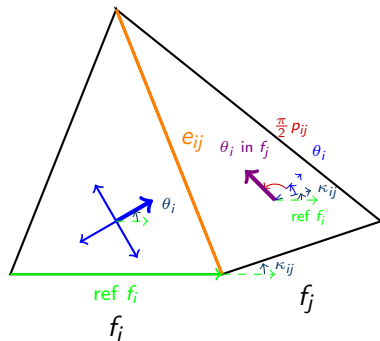
Mathematical Relationship

For adjacent triangles f_i and f_j , we can project θ_i into the f_j local coordinate system using:

$$\underbrace{\theta_i}_{\theta_i \text{ in } f_i} \mapsto \underbrace{\theta_i + \kappa_{ij} + \frac{\pi}{2} p_{ij}}_{\theta_i \text{ in } f_j}$$

Components:

- θ_i : frame angle in f_i
- κ_{ij} : geometric rotation
- $\frac{\pi}{2} p_{ij}$: symmetry jump
- θ_j : resulting angle in f_j



The main direction from f_i (solid blue) appears in f_j 's coordinate system (solid purple) after applying the transformation $\theta_i + \kappa_{ij} + \frac{\pi}{2} p_{ij}$ with $p_{ij} = 1$.

Smoothness Energy

Energy Definition

The smoothness energy measures misalignment between adjacent frames:

$$E_{\text{smooth}} = \sum_{e_{ij}} (\theta_i - \theta_j)^2 = \sum_{e_{ij}} \underbrace{(\theta_i + \kappa_{ij} + \frac{\pi}{2} p_{ij} - \theta_j)^2}_{\theta_i \text{ in } f_j} \stackrel{\text{ideal}}{=} 0$$

Goal: Minimize E_{smooth} to make frames as aligned as possible with respect to user constraints.

Difficulty: Mixed-integer problem since $p_{ij} \in \mathbb{Z}$ (integer) and $\theta_i \in \mathbb{R}$ (continuous).

Optimization Problem

Mixed Integer Problem

For a domain with n faces and m period jumps, we want to solve:

$$\min_{\vec{\theta} \in \mathbb{R}^n, \vec{p} \in \mathbb{Z}^m} E_{\text{smooth}}(\vec{\theta}, \vec{p}) = \min_{\vec{\theta} \in \mathbb{R}^n, \vec{p} \in \mathbb{Z}^m} \sum_{e_{ij}} \left(\theta_i + \kappa_{ij} + \frac{\pi}{2} p_{ij} - \theta_j \right)^2$$

Continuous relaxation: Taking derivatives with respect to variables:

$$\frac{\partial E}{\partial p_{ij}} = 2 \left(\theta_i + \kappa_{ij} + \frac{\pi}{2} p_{ij} - \theta_j \right) \cdot \frac{\pi}{2} = \pi \left(\theta_i + \kappa_{ij} + \frac{\pi}{2} p_{ij} - \theta_j \right) = 0$$

$$\frac{\partial E}{\partial \theta_i} = \sum_{j \in N(i)} 2 \left(\theta_i + \kappa_{ij} + \frac{\pi}{2} p_{ij} - \theta_j \right) = 0$$

Matrix Formulation

Linear system $Ax = b$ **for the continuous relaxation:** Ordering the variables as $x = (\mathbf{p}^T, \theta^T)^T$:

$$\begin{pmatrix} \frac{\pi}{2}I & M^T \\ \frac{\pi}{2}M & L \end{pmatrix} \begin{pmatrix} \mathbf{p} \\ \theta \end{pmatrix} = \begin{pmatrix} -\kappa_{\text{edge}} \\ -\kappa_{\text{vert}} \end{pmatrix}$$

Here, M is the edge-face incidence matrix and $L = MM^T$ is the graph Laplacian.

Explicitly for edge k (connecting $i \rightarrow j$) and face i :

$$\left(\begin{array}{ccc|ccc} \ddots & 0 & 0 & \dots & \vdots & \vdots \\ 0 & \frac{\pi}{2} & 0 & \dots & 1 & -1 & \dots \\ 0 & 0 & \ddots & & \vdots & \vdots \\ \hline \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \dots & \frac{\pi}{2} & \dots & \dots & \text{deg}(i) & -1 & \dots \\ \vdots & \vdots & \vdots & & \vdots & \ddots & \end{array} \right) \begin{pmatrix} \vdots \\ p_k \\ \vdots \\ \hline \vdots \\ \theta_i \\ \theta_j \\ \vdots \end{pmatrix} = \begin{pmatrix} \vdots \\ -\kappa_k \\ \vdots \\ \hline \vdots \\ -\sum \kappa_{ij} \\ \vdots \end{pmatrix}$$

Algorithm: Greedy Rounding

- 1 **Initialize:** Solve continuous relaxation ($p_{ij} \in \mathbb{R}$) for global solution x^0 .
 - 2 **Iterate** until all p_{ij} are integers:
 - Find variable p_k with smallest rounding error:
 $k = \arg \min_i |p_i - \text{round}(p_i)|$.
 - **Round and Lock:** $p_k \leftarrow \text{round}(p_k)$.
 - **Update:** Re-solve for remaining free variables $(\theta, p_{\text{free}})$ to compensate for the change.
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Why Greedy?

- **Greedy Strategy:** Rounding the "safest" variable first allows the system to *relax* the remaining θ variables to compensate for the rounding error.

Efficiency (Adaptive):

- Full re-solving is too slow.
- Use **Local Gauss-Seidel:** Only update variables locally dependent on the locked p_k .